

CHAPTER III THE STRAIGHT LINE

Art. 3.1 Two equations of the first degree in x, y, z together represent

$$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2 \quad (1)$$

which is the line of intersection of the two given planes

$$a_1x + b_1y + c_1z + d_1 = 0 \quad (2)$$

$$\text{and} \quad a_2x + b_2y + c_2z + d_2 = 0 \quad (3)$$

and we know that planes (2) and (3) intersect in a straight line.



Art. 3.2 Parametric equation of a straight line

Let the direction cosines of the line L be l, m, n and let $P_1(x_1, y_1, z_1)$ be a point on the line. Then L is the locus of $P(x, y, z)$ which is moving so that

$$x - x_1 = r \cos \alpha = rl, \quad y - y_1 = r \cos \beta = rm \quad \text{and} \quad z - z_1 = r \cos \gamma = rn.$$

$$\text{Therefore, } x = x_1 + rl, \quad y = y_1 + rm \quad \text{and} \quad z = z_1 + rn, \quad (4)$$

where the parameter r represents the variable length PP_1 .

If a, b, c be the direction ratios of L , then (4) can be written as

$$x = x_1 + at, \quad y = y_1 + bt \quad \text{and} \quad z = z_1 + ct, \quad (5)$$

where t does not represent the distance of the variable length PP_1 .

The equation (5) represents the coordinates of any point on the line L by giving different values to t .

Art. 3.3 (a) Symmetrical form of a straight line in terms of direction cosines.

To find the equation of a straight line through a given point in a given direction.

Let the given point be (x_1, y_1, z_1) and the direction be defined by the direction cosines l, m, n .

Let $P(x, y, z)$ be any point on the line and let $|PP_1| = r$. Then from Art. 29, we have

$$x = x_1 + rl, \quad y = y_1 + rm \quad \text{and} \quad z = z_1 + rn.$$

So that for all points (x, y, z) on PP_1 ,

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r \quad (6)$$

$$\text{i.e.,} \quad \frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} \quad (7)$$

is the symmetrical form of the line.

Art. 3.3 (b) Symmetrical form of a straight line in terms of direction ratios

If a, b, c be the direction ratios of the line, then we have

$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}} = \frac{a}{\sqrt{\sum a^2}}, \quad m = \frac{b}{\sqrt{\sum a^2}} \quad \text{and} \quad n = \frac{c}{\sqrt{\sum a^2}}.$$

$$\text{Then equation (7) can be written as} \quad \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} \quad (8)$$

Corollary. If this line is perpendicular to one of the coordinate axes, then (8) takes the form

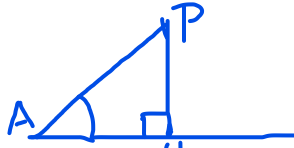
$$\begin{aligned} x = x_1, \quad \frac{y - y_1}{b} &= \frac{z - z_1}{c} \quad (\text{when the line is perpendicular to } x\text{-axis}) \\ y = y_1, \quad \frac{x - x_1}{a} &= \frac{z - z_1}{c} \quad (\text{when the line is perpendicular to } y\text{-axis}) \\ z = z_1, \quad \frac{x - x_1}{a} &= \frac{y - y_1}{b} \quad (\text{when the line is perpendicular to } z\text{-axis}). \end{aligned}$$

Art. 3.4 Line through two given points

If the coordinates of P, Q be $(x_1, y_1, z_1), (x_2, y_2, z_2)$, then the direction ratios of PQ are proportional to $x_2 - x_1, y_2 - y_1, z_2 - z_1$ and therefore the equations of PQ are

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1} \quad (9)$$

Art. 3.5 Distance of a point from a line.



Let the length of the perpendicular distance from a point $P(x_1, y_1, z_1)$ to the line

$$\frac{x - a}{l} = \frac{y - b}{m} = \frac{z - c}{n} \quad (10)$$

be p . If $A(a, b, c)$ be any point on the given line and if N is the foot of the perpendicular from P on it, then we can write

$$PN^2 = AP^2 - AN^2 \quad (11)$$

But $AP^2 = (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2$ and

$$\begin{aligned} AN &= \text{projection of } AP \text{ on the line } AN \\ &= l(x_1 - a) + m(y_1 - b) + n(z_1 - c) \end{aligned}$$

provided that l, m, n are the actual direction cosines of AP .

$$\begin{aligned} \text{Therefore, } PN^2 &= (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2 - \{l(x_1 - a) + m(y_1 - b) + n(z_1 - c)\}^2 \\ &= \{(x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2\}(l^2 + m^2 + n^2) - \{l(x_1 - a) + m(y_1 - b) + n(z_1 - c)\}^2 \\ \Rightarrow p^2 &= \{l(y_1 - b) - m(x_1 - a)\}^2 + \{m(z_1 - c) - n(y_1 - b)\}^2 + \{n(x_1 - a) - l(z_1 - c)\}^2 \\ &= \begin{vmatrix} x_1 - a & y_1 - b \\ l & m \end{vmatrix}^2 + \begin{vmatrix} y_1 - b & z_1 - c \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1 - c & x_1 - a \\ n & l \end{vmatrix}^2. \end{aligned}$$

Art. 3.6 Angle between a line and a plane

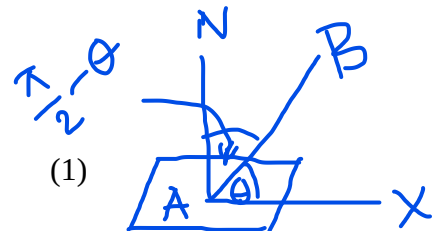
Let the equation of the line be

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$$

and the plane be

$$ax + by + cz + d = 0$$

(2)



Let θ be the line between the line AB and the plane (2). Then $\frac{\pi}{2} - \theta$ is the angle between the line and normal AN to the plane.

Direction ratios of AN are a, b, c and direction ratios of the line are l, m, n . Then

$$\cos\left(\frac{\pi}{2} - \theta\right) = \frac{al + bm + cn}{\sqrt{a^2 + b^2 + c^2} \sqrt{l^2 + m^2 + n^2}}$$

$$\text{i.e., } \sin \theta = \frac{al + bm + cn}{\sqrt{a^2 + b^2 + c^2} \sqrt{l^2 + m^2 + n^2}}$$

gives the angle between the line (1) and the plane (2).

Corollary: (1) If the line (1) is parallel to the plane (2) (i.e., if the line (1) is perpendicular to the normal AN to (2)), then $al + bm + cn = 0$.

(2) If the line (1) is perpendicular to the plane (2) (i.e., if the line (1) is parallel to the normal AN to (2)), then $\frac{a}{l} = \frac{b}{m} = \frac{c}{n}$.

Art. 3.7 Conditions for which a plane contains a line

Find the conditions that the line

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} \quad (1)$$

may lie in the plane $ax + by + cz + d = 0$ (2)

If the line (1) lies in the plane (2), then for all values of r , the point $(x_1 + lr, y_1 + mr, z_1 + nr)$ should lie in the plane (2).

Therefore, $a(x_1 + lr) + b(y_1 + mr) + c(z_1 + nr) + d = 0$

$$\Rightarrow r(al + bm + cn) + (ax_1 + by_1 + cz_1 + d) = 0 \quad (3)$$

So the conditions are

$$al + bm + cn = 0 \quad (4)$$

$$\text{and } ax_1 + by_1 + cz_1 + d = 0 \quad (5)$$

Condition (4) shows that is parallel to the plane and condition (5) shows that the point (x_1, y_1, z_1) on the line lies in the plane.

Note: The general equation of any plane containing the line (1) is

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

where $al + bm + cn = 0$.

Art. 3.7 Coplanar lines

Find the condition that the lines whose equations are given in symmetrical form may intersect and determine the equation of the plane in which they lie.

Let the lines are

$$\frac{x - x_1}{l_1} = \frac{y - y_1}{m_1} = \frac{z - z_1}{n_1} \quad (1)$$

$$\text{and } \frac{x - x_2}{l_2} = \frac{y - y_2}{m_2} = \frac{z - z_2}{n_2} \quad (2)$$

The equation of the plane passing through (x_1, y_1, z_1) is

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0 \quad (3)$$

where $al_1 + bm_1 + cn_1 = 0$ (4)

If (2) lies on (3), the point (x_2, y_2, z_2) lies on it and the line is perpendicular to the normal to the plane (3). In this case, we have

$$a(x_2 - x_1) + b(y_2 - y_1) + c(z_2 - z_1) = 0 \quad (5)$$

where $al_2 + bm_2 + cn_2 = 0$ (6)

Eliminating a, b, c from (5), (4) and (6), we get

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0 \quad (7)$$

which is the required condition of coplanarity of two lines.

Again, eliminating a, b, c from (3), (4) and (6), we get

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0 \quad (8)$$

which is the equation of the plane containing (1) and (2).

Corollary: If the equations of the planes are

$$a_1x + b_1y + c_1z + d_1 = 0, \quad a_2x + b_2y + c_2z + d_2 = 0,$$

$$a_3x + b_3y + c_3z + d_3 = 0, \quad \text{and} \quad a_4x + b_4y + c_4z + d_4 = 0,$$

then the condition of coplanarity is

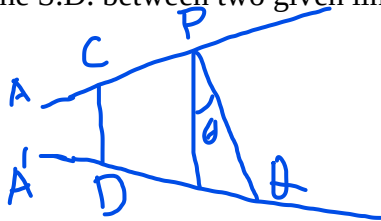
$$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix} = 0.$$

Art. 3.8 Shortest Distance

Definition

Lines which do not intersect or equivalently, the lines which do not lie in the same plane, are called *nonintersecting lines*. The straight line which is perpendicular to each of these lines is called the *line of shortest distance* (S.D.) and the length of this line is called the *length of the line* of S.D.

Art. Find the S.D. between two given lines and also find the equations of the line of S.D.



Here, $CD = \text{projection of } PQ \text{ on the line } CD = PQ \cos \theta$, where θ is the angle between PQ and CD .

Since $\cos \theta < 1$, PQ is always greater than CD . Thus, CD being always less than any distance between two lines is the line of S.D.

Let the equation of the given lines be

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1} \quad (1)$$

and
$$\frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2} \quad (2)$$

Let $A(x_1, y_1, z_1)$ be the point on line (1) and let $A'(x_2, y_2, z_2)$ be the point on the line (2).

Now suppose that α, β, γ be the direction cosines of CD which is perpendicular to both the lines.

Then we have $l_1\alpha + m_1\beta + n_1\gamma = 0 \quad (3)$

and $l_2\alpha + m_2\beta + n_2\gamma = 0 \quad (4)$

So that
$$\frac{\alpha}{m_1n_2 - m_2n_1} = \frac{\beta}{n_1l_2 - n_2l_1} = \frac{\gamma}{l_1m_2 - l_2m_1} = \frac{\sqrt{\alpha^2 + \beta^2 + \gamma^2}}{\sqrt{\sum(m_1n_2 - m_2n_1)}} = \frac{1}{\sin \theta}.$$

Hence, $\alpha = \frac{m_1n_2 - m_2n_1}{\sin \theta}, \beta = \frac{n_1l_2 - n_2l_1}{\sin \theta},$ and $\gamma = \frac{l_1m_2 - l_2m_1}{\sin \theta}.$

So, the shortest distance, S.D. = projection of the line joining $A(x_1, y_1, z_1)$ and $A'(x_2, y_2, z_2)$ on the line of S.D. whose direction cosines are α, β, γ .

Therefore, S.D. = $(x_2 - x_1)\alpha + (y_2 - y_1)\beta + (z_2 - z_1)\gamma$

$$\begin{aligned} &= \frac{1}{\sin \theta} \{(x_2 - x_1)(m_1n_2 - m_2n_1) + (y_2 - y_1)(n_1l_2 - n_2l_1) + (z_2 - z_1)(l_1m_2 - l_2m_1)\} \\ &= \frac{1}{\sin \theta} \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} \end{aligned}$$

Now, the equation of the plane containing (1) and parallel to the line of S.D. is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ \alpha & \beta & \gamma \end{vmatrix} = 0 \quad (5)$$

Now, the equation of the plane containing (2) and parallel to the line of S.D. is

$$\begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ l_2 & m_2 & n_2 \\ \alpha & \beta & \gamma \end{vmatrix} = 0 \quad (6)$$

Therefore the equation of the line of S.D. is given by

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ \alpha & \beta & \gamma \end{vmatrix} = 0 = \begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ l_2 & m_2 & n_2 \\ \alpha & \beta & \gamma \end{vmatrix} \quad (7)$$

Worked Out Problems

1. Show that the lines $3x - 2y + 13 = 0 = y + 3z - 26$ and $\frac{x+4}{5} = \frac{y-1}{-3} = \frac{z-3}{1}$ are perpendicular.

Solution. Given $3x - 2y + 13 = 0 = y + 3z - 26$ (1)

$$\text{and } \frac{x+4}{5} = \frac{y-1}{-3} = \frac{z-3}{1} \quad (2)$$

The drs of the first line are

$$\begin{array}{ccccc} 3 & -2 & 0 & 3 & -2 \\ 0 & 1 & 3 & 0 & 1 \end{array}$$

$\Rightarrow -6, -9, 3$ or $2, 3, -1$ and those of (2) are $5, -3, 1$.

Since $(2)(5) + (3)(-3) + (-1)(1) = 0$, the given lines are perpendicular.

2. Find the angle between the line $2x + 4y - 2z + 3 = 0 = 4x - 2y + 6z + 5$ and the plane $5x - 4y + 3z - 5 = 0$.

Solution. The angle between the line and the plane means the angle between the line and the normal to the plane.

Given $2x + 4y - 2z + 3 = 0 = 4x - 2y + 6z + 5$ (1)

and $5x - 4y + 3z - 5 = 0$ (2)

The drs of the line are

$$\begin{array}{ccccc} 2 & 4 & -2 & 2 & 4 \\ 4 & -2 & 6 & 4 & -2 \end{array}$$

$\Rightarrow 20, -20, -20$ or $1, -1, -1$ and the drs of the normal to the plane (2) are $5, -4, 3$.

Let θ be the angle between the line and the normal to the plane (2).

$$\text{Then } \cos \theta = \pm \frac{(1)(5) + (-1)(-5) + (-1)(3)}{\sqrt{1+1+1}\sqrt{25+25+9}} = \pm \frac{6}{\sqrt{150}}$$

$$\Rightarrow \theta = \cos^{-1} \left(\pm \frac{6}{\sqrt{150}} \right).$$

3. Write the line $x + y + z + 1 = 0 = 4x + y - 2z + 2$ in symmetrical form.

Solution. Given $x + y + z + 1 = 0 = 4x + y - 2z + 2$ (1)

The drs of the given line are

$$\begin{array}{ccccc} 1 & 1 & 1 & 1 & 1 \\ 4 & 1 & -2 & 4 & 1 \end{array} \Rightarrow -3, 6, -3 \text{ or } -1, 2, -1.$$

Putting $z = 0$ in (1), we get $x + y + 1 = 0$ and $4x + y + 2 = 0$. So the actual direction cosines of (1) are $-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}$.

Solving them, we get $x = -\frac{1}{3}$ and $y = -\frac{2}{3}$. So $(-\frac{1}{3}, -\frac{2}{3}, 0)$ is the point on the given line with direction ratios $-1, 2, -1$.

Hence, in symmetrical form, the equations of the line are

$$\frac{x + \frac{1}{3}}{-1} = \frac{y + \frac{2}{3}}{2} = \frac{z - 0}{-1}.$$

4. Find the equation of the line perpendicular from the origin to the line

$$x + 2y + 3z + 4 = 0 = 2x + 3y + 4z + 5.$$

Also find the coordinates of the foot of the perpendicular.

Solution. Given $x + 2y + 3z + 4 = 0 = 2x + 3y + 4z + 5$ (1)

The drs of the given line are

$$\begin{array}{ccccc} 1 & 2 & 3 & 1 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{array} \Rightarrow -1, 2, -1 \text{ or } 1, -2, 1.$$

Putting $z = 0$ in (1), we get $x + 2y + 4 = 0$ and $2x + 3y + 5 = 0$.

Solving them, we get $x = 2$ and $y = -3$. So $(2, -3, 0)$ is a point on the given line and it can be written in the symmetrical form as

$$\frac{x - 2}{1} = \frac{y + 3}{-2} = \frac{z - 0}{1} = r \text{ (say).}$$

If N is any point on the line, then the coordinates of N are $(r + 2, -2r - 3, r)$. If N is the foot of the perpendicular from $P(0, 0, 0)$ on the given line. So the drs of PN are $r + 2 - 0, -2r - 3 - 0, r - 0$, i.e., $r + 2, -2r - 3, r$.

Since PN and the given line are perpendicular, then

$$1(r + 2) - 2(-2r - 3) + 1r = 0 \Rightarrow r = -\frac{4}{3}$$

So the coordinates of N are $(\frac{2}{3}, -\frac{1}{3}, -\frac{4}{3})$ and the direction ratios of PN are $2, -1, -4$.

Thus the equations of the line perpendicular from P to the given line are

$$\frac{x - 0}{2} = \frac{y - 0}{-1} = \frac{z - 0}{-4}.$$

5. Show that the lines

$$\frac{x+5}{3} = \frac{y+4}{1} = \frac{z-7}{-2} \text{ and } 3x+2y+z-2=0 = x-3y+2z-13$$

are coplanar. Also find the equation to the plane in which they lie.

Solution. Given
$$\frac{x+5}{3} = \frac{y+4}{1} = \frac{z-7}{-2} \quad (1)$$

$$3x+2y+z-2=0 = x-3y+2z-13 \quad (2)$$

Any plane passing through (2) is

$$3x+2y+z-2+k(x-3y+2z-13)=0$$

$$\Rightarrow (3+k)x + (2-3k)y + (1+2k)z + (-2-13k) = 0 \quad (3)$$

If (1) and (3) are parallel, we have

$$3(3+k) + 1(2-3k) - 2(1+2k) = 0 \Rightarrow k = \frac{9}{4}$$

Hence the equation of the plane containing the second line and parallel to the first line is

$$(3+\frac{9}{4})x + (2-\frac{27}{4})y + (1+\frac{9}{2})z + (-2-\frac{117}{4}) = 0$$

$$\Rightarrow 21x - 19y + 22z - 125 = 0$$

which passes through the point $(-5, -4, 7)$ and so this contains the first line also.

Hence the given lines are coplanar and they in the plane $21x - 19y + 22z - 125 = 0$.

6. Find the equation of the line perpendicular to the lines

$$\frac{x-1}{1} = \frac{y-1}{2} = \frac{z+2}{3} \text{ and } \frac{x+2}{2} = \frac{y-5}{-1} = \frac{z+3}{2}$$

and passing through their intersection.

Solution. Let $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z+2}{3} = r \Rightarrow x=1+r, y=1+2r, z=-2+3r$

Again, let $\frac{x+2}{2} = \frac{y-5}{-1} = \frac{z+3}{2} = R \Rightarrow x=-2+2R, y=5-R, z=-3+2R$

If the given lines meet, then $1+r = -2+2R, 1+2r = 5-R, -2+3r = -3+2R$

Solving them, we get $r=1$ and $R=2$.

Hence the coordinates of the point of intersection are $(2, 3, 1)$.

Let l, m, n be the dcs of the required line. Then we have $l+2m+3n=0$ and $2l-m+2n=0$

whence $\frac{l}{7} = \frac{m}{4} = \frac{n}{-5}$.

So the drs of the required line are $7, 4, -5$ and the equations of the required lines are

$$\frac{x-2}{7} = \frac{y-3}{4} = \frac{z-1}{-5}.$$

7. Find the equation of the line drawn parallel to

$$\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$$

so as to intersect

$$9x + y + z + 4 = 0 = 5x + y + 3z$$

$$x + 2y - 3z - 3 = 0 = 2x - 5y + 3z + 3.$$

8. Find the distance of the point $(-1, -5, -10)$ from the point of intersection of the line

$$\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$$

and the plane $x - y + z = 5$.

9. Find the length and equation of the line of shortest distance between the lines

$$\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1} \text{ and } \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}.$$

Find also the points in which it meets the given lines.

10. Find the length and equation of the line of shortest distance between

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \text{ and } \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}.$$

Determine whether the lines are coplanar or not.

11. Suppose $\frac{x}{2} = \frac{y}{-3} = \frac{z}{6}$ be the equation of the line AB . Through $P(1, 2, 5)$, PN is drawn

perpendicular to AB and PQ is drawn parallel to the plane $3x + 4y + 5z = 0$ to meet AB in Q . Find the equation of PN and PQ .

Solution. Let $\frac{x}{2} = \frac{y}{-3} = \frac{z}{6} = r \Rightarrow x = 2r, y = -3r, z = 6r$, and so the coordinates of N are $(2r, -$

$3r, 6r)$ and the drs of PN are $2r - 1, -3r - 2, 6r - 5$.

Since PN is perpendicular to AB , we have $2(2r - 1) - 3(-3r - 2) + 6(6r - 5) = 0$ which gives

$r = \frac{26}{49}$. So the coordinates of N are $\left(\frac{52}{49}, -\frac{78}{49}, \frac{156}{49}\right)$.

Hence the equations of PN are

$$\frac{x-1}{1-\frac{52}{49}} = \frac{y-2}{2+\frac{78}{49}} = \frac{z-5}{5-\frac{156}{49}} \Rightarrow \frac{x-1}{-3} = \frac{y-2}{176} = \frac{z-5}{89}$$

Again, let $\frac{x}{2} = \frac{y}{-3} = \frac{z}{6} = R \Rightarrow x = 2R, y = -3R, z = 6R$, and so the coordinates of Q are $(2R, -$

$3R, 6R)$ and the drs of PQ are $2R - 1, -3R - 2, 6R - 5$.

Since PQ is parallel to $3x + 4y + 5z = 0$, we have

$$3(2R - 1) + 4(-3R - 2) + (6R - 5) = 0 \Rightarrow R = \frac{3}{2}$$

So the coordinates of Q are $(3, -\frac{9}{2}, 9)$ and finally, the equations of PQ are

$$\frac{x-1}{1-3} = \frac{y-2}{2+\frac{9}{2}} = \frac{z-5}{5-9} \Rightarrow \frac{x-1}{-4} = \frac{y-2}{13} = \frac{z-5}{-8}.$$

12. Through the point $P(-1, 1, 2)$ a line is drawn parallel to the line of intersection of the planes $x - 2y + z = 3$ and $x + 6y - 5z = 0$. This line cuts the plane $x - 3y + 2z = 2$ in Q. Find the coordinates of Q and the equation of the line PQ.

Solution. The line of intersection of the given planes are

$$x - 2y + z - 3 = 0 = x + 6y - 5z \quad (1)$$

The drs are

$$\begin{array}{ccccc} 1 & -2 & 1 & 1 & -2 \\ 1 & 6 & -5 & 1 & 6 \end{array} \Rightarrow 4, 6, 8 \text{ or } 2, 3, 4.$$

So the drs of the line parallel to (1) are 2, 3, 4.

The equations of the line through $P(-1, 1, 2)$ are

$$\frac{x+1}{2} = \frac{y-1}{3} = \frac{z-2}{4} = r, \text{ (say)} \quad (2)$$

Since (2) cuts the plane $x - 3y + 2z = 2$, then $(2r - 1) - 3(3r + 1) + 2(4r + 2) = 0 \Rightarrow r = 2$.

So the coordinates of Q are $(3, 7, 10)$.

Hence the equations of PQ are

$$\frac{x+1}{-1-3} = \frac{y-1}{1-7} = \frac{z-2}{2-10} \Rightarrow \frac{x+1}{2} = \frac{y-1}{3} = \frac{z-2}{4}.$$